Phase I System Presentation

Team EV

Members: Zichuo Wang, James Zuniga, Austin Aragon, Justus Jones, Young Kim

SCENARIO 1

Basic Step-by-Step Guidance

Feature: Basic step-by-step guidance | **User Group: Primary user**

AS-IS SCENARIO

- Stevie is attempting to reach his next classroom independently. He knows he must walk forward a few paces and turn left at the corridor corner.
- Lacking distance confirmation, he is unsure of the exact turning threshold. He guesses the turn prematurely, runs into the corridor corner, and sustains a minor head injury.

TO-BE SCENARIO

- THEIA prompts Stevie to enter or dictate his current starting point and intended destination.
- THEIA calculates an optimal, hazard-free route and provides clear cadence: *"Walk ahead 10 steps, then turn left."*
- Precise distance pacing prevents premature turns and protects Stevie from structural impacts.

SCENARIO 2

Off-Route Recovery & Emergency Escalation

Feature: Off-route recovery + emergency help

User Group: Primary user, caretaker, campus police

AS-IS SCENARIO

- Alfred heads toward the 3rd-floor elevator after an evening lab when construction barricades block his standard path.
- After detouring, Alfred becomes disoriented on the empty floor. Alfred calls caretaker Bella but can only state he is “somewhere on the 3rd floor.” Bella must search the entire floor manually to locate Alfred.

TO-BE SCENARIO

- Alfred tells THEIA “this way is blocked” and immediately receives an alternate path. When a missed beacon indicates deviation, THEIA warns Alfred that he may be off route and guides him back to last known beacon (Spark 233).
- If Alfred speaks “Help” or an unresponsive fall is detected, THEIA dispatches an automated SMS alert with his exact location to Bella, offering escalation to campus police.

SCENARIO 3

Multi-Floor Navigation by Elevator

Feature: Multi-floor navigation by elevator | **User Group: Primary user**

AS-IS SCENARIO

- Mark exits TFLO 239 carrying heavy equipment. He needs to reach TFLO 131 but cannot safely take the stairs under load.
- Without dedicated audio navigation to find the elevator, Mark makes 3 wrong turns in the corridor trying to reach the call button, finally relying on a student for directions and arriving 10 minutes late.

TO-BE SCENARIO

- Mark commands: *"THEIA, lead me to TFLO 131 via elevator."* The app generates the shortest accessible route with minimal turns directly to the elevator bank.
- Audio cues are used to direct Mark toward the elevator doors, and haptic pulses when arriving at the elevator confirm exact arrival.
- Upon reaching floor 1, THEIA auto-switches floor plans and continues seamless turn guidance to his classroom.

SCENARIO 4

Personal Settings & Navigation Modes

Feature: Personal settings + navigation modes | **User Group: Primary user, caretaker**

AS-IS SCENARIO

- Elena attempts to reach the cafeteria using standard audio maps with a rigid 3-second interval, which is drowned out by ambient noise.
- She wants to increase cue frequency and choose calmer hallways but cannot modify complex settings on her own device which requires her caretaker to reconfigure beforehand.

TO-BE SCENARIO

- Elena states, *"I'm heading to lunch at the cafeteria,"* and double-taps the screen to confirm.
- Elena says *"THEIA, louder"* and *"instructions every 10 seconds"*; THEIA confirms each adjustment aloud and gives instructions more often.
- THEIA selects a route that avoids areas the map marks as busy, like the main lobby, while emergency contacts set by her caretaker remain locked and secure.

Top Priority Analysis: Off-Route Recovery (Scenario 2)

WHO

Alfred as primary user, Bella as 1st secondary user (caretaker), with Campus Police as 2nd secondary user (backup).

WHAT

Off-route detection, guide back to last known location, and automated SMS emergency dispatch.

WHEN & WHERE

During any trip (especially after detours in empty buildings at night) across any mapped floor (e.g., Spark 233).

WHY TOP PRIORITY

Safety-critical and high impact, it provides contextual failsafe recovery that a white cane or service dog cannot deliver on it own.

HOW: Missed Beacon → "May be off route" Alert → Reroute or Guide to Last Beacon → User Says "Help" → Caretaker Gets Text Location

WHAT-IF EDGE CASES (LESSON 6 REQUIREMENT)

No beacon nearby? THEIA announces last confirmed location and guides user along the nearest wall.

Unresponsive fall? System initiates a 30-second countdown, then automatically sends location SMS.

Caretaker unresponsive? THEIA immediately prompts escalation to Campus Police dispatch.

False alarm triggered? User can verbally cancel within the alert window by saying *"Cancel"*.

Why THEIA? Comparative Analysis

A complementary tool to use along with a cane or service animal.

White Cane

Strengths: Inexpensive, reliable physical ground-level obstacle detection.

Limitations: Cannot provide context for indoor room numbers, floor plans, or multi-floor routing.

Service Animal

Strengths: Highly effective for moving around obstacles, drop-offs, and stairs.

Limitations: High training and care cost, cannot read room labels, navigate elevators, or send emergency SMS alerts.

Project THEIA

Combines hands-free voice + vibration guidance, room-level indoor navigation across multiple floors, and elevator waypoint tracking.

Safety Failsafe: Automated fall detection, off-route recovery, and direct SMS escalation to designated caretakers.